



**KISI-KISI / PETUNJUK
SELEKSI DAERAH CALON KOMPETITOR
ASEAN SKILLS COMPETITION (ASC) XII
TAHUN 2019
KEJURUAN IT SOFTWARE SOLUTION FOR
BUSINESS**

**DIREKTORAT BINA STANDARDISASI KOMPETENSI DAN PELATIHAN KERJA
DIREKTORAT JENDERAL PEMBINAAN PELATIHAN DAN PRODUKTIVITAS
KEMENTERIAN KETENAGAKERJAAN RI**

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PETUNJUK
SELEKDA 2019 ASEAN SKILLS COMPETITION XII
KEJURUAN IT SOFTWARE SOLUTION FOR BUSINESS

Pendahuluan

Test Project (TP) seleksi daerah ini dirancang untuk dapat mengevaluasi kemampuan peserta secara objektif dalam merancang dan membangun aplikasi untuk mendukung system informasi dalam perusahaan. Tes project terbagi dalam beberapa modul yang didasari evaluasi kemampuan terkait :

1. Kemampuan untuk melihat dan merancang proses bisnis yang tepat berbasis computer
2. Merancang database (ERD) yang diperlukan dalam pembangunan system (menggunakan SQL Server)
3. Merancang system untuk memberikan solusi terkait proses di dalam perusahaan
4. Melakukan coding / eksekusi terkait aplikasi yang ingin dibangun.

Pelaksanaan Test Project (TP) akan menggunakan PC (desktop) atau laptop dengan software yang sesuai. Kemampuan yang dinilai adalah 100% secara objektif. Software yang akan digunakan yaitu:

- Windows 10
- Microsoft Office 365 / office 2016 (untuk membaca soal)
- Visual Studio 2015 Community Edition / Visual Studio 2017 Community edition
- Microsoft SQL Server 2017 Express / Developer Edition
- Microsoft SQL Server Management Studio 2017 Express / Developer Edition

Kompetensi dan Ruang Lingkup

Test project kejuruan terdiri dari 4 modul yang menekankan pada kemampuan peserta untuk merancang solusi terkait kebutuhan database dan kebutuhan system informasi dalam kasus yang ada. Dari rancangan database yang dibuat akan dieksekusi menggunakan SQL Server. Kemudian di rancang system untuk menyelesaikan baik untuk master maupun transaksi.

Modul 1 : Merancang database sesuai kasus yang di berikan dan eksekusinya, membuat login, dan pembagian hak akses user

Modul 2 : Merancang dan coding terkait master yang ada sesuai kasus.

Modul 3 : Merancang dan coding terkait transaksi sesuai kasus.

Contoh perancangan database dan contoh pembuatan aplikasi ada di file terpisah (contoh soal kisi-kisi)

Hal yang perlu dipelajari :

1. Datagridview
2. Connect visual studio to sql server
3. Select , join table, insert, update, delete
4. Aggregate function
5. Upload image
6. Fungsi – fungsi string
7. Menambahkan component windows form (button, label, dsb) menggunakan code

Waktu Seleksi

Material Test Project yang diujikan memiliki durasi sebagai berikut

Modul	Durasi
Modul 1	3 jam
Modul 2	3 jam
Modul 3	3 jam
Modul 4	3 jam
Total	12 jam

Apabila peserta mampu menyelesaikan soal lebih cepat dari waktu yang ditentukan, peserta diperkenankan untuk meninggalkan area lomba tetapi hal ini (waktu) tidak masuk dalam poin penilaian.

Jadwal pelaksanaan Seleksi direncanakan sbb :

Hari	Pagi	Siang
Hari 1	• Pembukaan • Technical Briefing • Sosialisasi dan diskusi Test Project	• Kompetisi - Modul 1
Hari 2	Kompetisi - Modul 2	• Kompetisi - Modul 3
Hari 3	Kompetisi – Modul 4	Penutupan

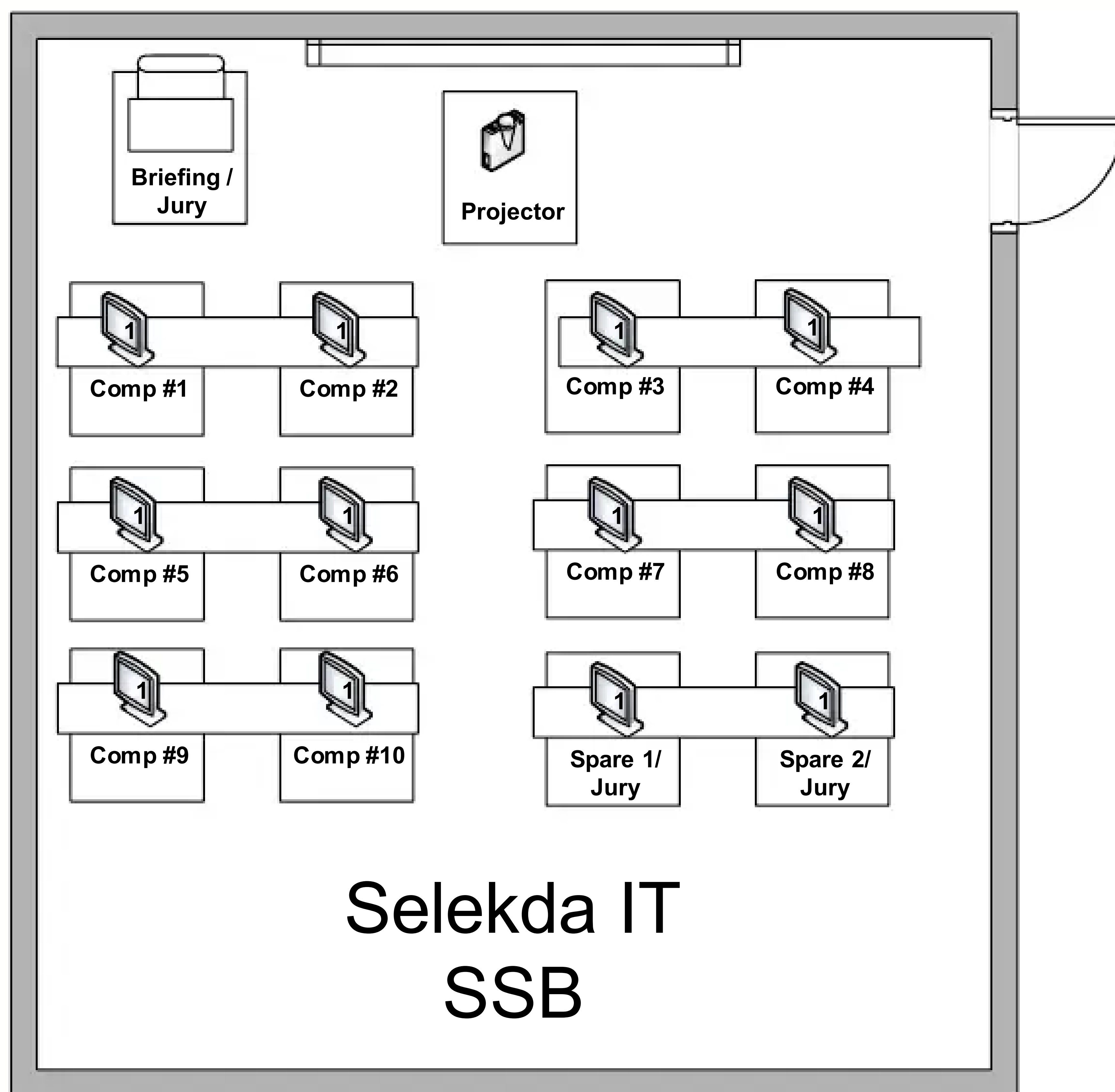
*Technical Briefing: Schedule, Rules & Procedure, MTP, Equipment & Software, Assessment Criteria

Skema Penilaian

Untuk setiap soal yang diberikan akan diberikan marking scheme sesuai permintaan soal. Marking secara detail akan di sosialisasikan bersamaan dengan sosialisasi test project.

Layout

Berikut adalah ilustrasi layout untuk 10 + 3 set meja untuk 10 peserta dan 3 spare / juri.



Gambar 1. Ilustrasi Layout ruang selekda untuk 10 kompetitor

Panitia

Peralatan dan material untuk seleksi mengikuti data alat dan bahan yang dilampirkan terpisah.

Peserta

Peralatan dan Material untuk perlombaan akan disediakan panitia. Peserta diijinkan membawa dan menggunakan peralatan pribadi seperti mouse dan atau keyboard non-programmable

Lain-lain

Peserta yang terlambat memasuki ruang lomba tidak diberikan perpanjangan waktu.

Sebelum kompetisi, akan dilakukan familiarisasi alat dan software sekitar 15-30 menit. Peserta wajib memeriksa kelengkapan dan fungsionalitas alat dan bahan yang digunakan untuk menjamin kelancaran pengerjaan test project.

Segala bentuk materi pembelajaran dan alat komunikasi dilarang dibawa ke area kompetisi.

Terdapat modul yang disajikan dalam format gambar atau diagram, bacalah instruksi modul yang diberikan dengan seksama.

Peserta dapat mengajukan pertanyaan ke juri terkait soal namun pertanyaan yang mengarah ke jawaban tidak akan dijawab oleh juri.

Kontak koordinasi persiapan

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Contoh Soal

Pemuda Theme Park is a playing ground that provides games and attraction for both children and adults. The strategic location makes Pemuda Theme Park as the most visited place by local and foreign tourists. With nearly 20 games and 10 attractions that lasted on average of 10 minutes for each game and 15 minutes for each attraction, Pemuda Theme Park are always full of visitors especially on weekend days (Saturday and Sunday).

Every games and attractions are operated by 4 staffs to support the operational, and making sure that all visitors will play in safe and proper order. To support game development, Pemuda Theme Park has Information System (software) they developed in-house. This software also used to support the staff in Pemuda Theme Park running a more effective operation.

The detail business processes of Pemuda Theme Park are explained below:

1. Ticketing Process

All visitors are required to buy entrance ticket to enjoy all games and attraction inside Pemuda Theme Park. The pricing for the tickets are as follows:

Weekday		Weekend	
Children < 13 years old	Rp 40.000	Children < 13 years old	Rp 60.000
Adult	Rp 50.000	Adult	Rp 80.000

The tickets purchased are only valid for one day use, on the day of purchase.

Pemuda Theme Park also offers membership card for that valid for desired period that can be used to enter the theme park also, with the pricing below:

Membership Plan	Price
3 months period	Rp. 200.000,-
6 months period	Rp. 300.000,-
12 months period	Rp. 500.000,-

The visitors who have bought the tickets or using the membership card will be stamped in hand indicating that they can enter the theme park to enjoy the games and attractions. These processes are supported using ticketing program, so the owner of Pemuda Theme Park can easily monitor their income using one-or-two clicks away in a real-time calculation.

If the member has been expired, the member is able to do the renewal. When renewing, the member can choose the specific membership plan. If the visitors come as a group, the visitors who like to get the membership plan need to register separately. So, the transaction will be divided into 2 parts which are the membership registration and the one entry registration.

2. Staff Scheduling

Every games and attraction need 1-4 employees to handling the operational process. Each day, employees will be allocated to operate everygame or attraction based on their working shift. There are 2 (two) shift for the employee:

- Morning shift (07.00 – 13.00) and
- Afternoon shift (13.00-19.00),

Pemuda Theme Park also use scheduling program to help them allocating the staff. Every week, the staff allocated for each games / attraction will be rolled, so that each of the games may have be operated by different staff. After the owner finish allocating the schedule, they will make scheduling time tables so all employee can look up their schedule for current week.

Here are the samples of the employee scheduling time table:

Games Shift	Niagara	Halilintar	Kicir-kicir	Arung Jeram	Tornado
Morning	Meity, Rudy, Desi	Budi, Andi, Eva	Renny, Vania, Yonathan	Rere, Gina, Anti	Andra, Lenny, Eka
Night	Karin, Santi, Deva	Reza, Toni, Rio	Vero, Ferdy, Juan	Joni, Ibnu, Tira	Laras, Geo, Didit

Pemuda Theme Park staff needs to clock their attendance in the system everyday by using the attendance terminal provided in the Owner Office room, whether it is a clock-in or clock-out attendance. This data will be used in payroll calculation, including if there is overtime for certain employees.

3. Payroll Process

The employee monthly salary is calculated using the working hour's data of each employee from their attendance. Every employee salary rate is determined as follows:

Type of Salary	Time Period	Rate
Daily Salary	Per one day full working hours (6 hours)	Rp. 200.000,-
Overtime Salary	Per overtime	Rp. 200.000,-
Monthly Bonus	Per one month full of attendance (full working days / month)	Rp. 300.000,-
Late Charge	Per late clock-in attendance / day	Rp. 5.000,-

Each month, the system will generate the amount of payment for each employee. Total amount of salary every employee get are calculated using formula below:

Work Salary = (Total Daily Salary + Total Overtime Salary) – Total Late Charge

If the employee coming for full work days per month, they will get additional Monthly Bonus, so the calculation will be as follow (take notes Monthly Bonus only acquired by employee who are eligible, and not all employee):

$\text{Total Salary} = \text{Work Salary} + \text{Monthly Bonus}$

Every month, the owner of Pemuda Theme Park want to monitor the payroll report for all employee. Make sure the owner of Pemuda Theme Park able to see the report with complete and detailed data in a real-time process.

4. Substitute Schedule / Overtime

When an employee is absent (not be able to do the work on that day), other employee can be sign up for the job. The employee must be available on that shift. Also, the shift will be counted as an overtime shift. When the substitute employee is been signed up, they also need to do the clock-in and clock-out on that day.

E.g. Meity have a morning schedule on Monday, but she can't be able to do her job on that day. Fortunately, Karin and Eva are willing to do Meity's shift on that day. The problem is Eva also have schedule on the same day in different game, so she is not possible to get the job. Finally, Karin who not have morning shift on that day will be signed up for Meity's shift and it is also counted as overtime shift for Karin.

5. Attendance Process

Each employee must do the absent manually in attendance sheet by writing clock-in, clock-out and signature each day. The owner will input the data from attendance sheet to the system.

6. Report Process

The owner can generate 2 kind of reports for each month, those are:

- Salary Report

This report will show information about salary of each staff and summary of the month.

The information about each staff per month which must be shown are EmployeeID, Name, Total Attendance, total Lateness, total overtime and total salary.

The information about summary of the month that must be shown are total salary of all staff, total lateness of all staff and total staff.

- ticketing report

This report will show information about total ticketing sales in specific chosen month. The information that must be shown are date, quantity of the ticket adult, quantity of the ticket child and quantity of the ticket member (free ticket)

After 10 years operating, it is the time to improve the software that had been used for years because the software cannot be running in the current operating system anymore. As a senior software developer, you are required to develop the new software to replace the old software in Pemuda Theme Park. Concerning Pemuda Theme Park business process above, please develop the new software to support the business process!

Session 1: Designing Information System including the Database

A good information system always comes from a good system design, in this development phase you are required to design the database and the user interface of the information system to support Pemuda Theme Park business process as explained before with the following criteria:

1. Entity Relationship Diagram

- a. Ensure the ERD developed can store all the information required in the system, with minimum 5 master table, and 5 transaction table
- b. Your designed ERD should be able to store data of Pemuda Theme Park core process, with concerning the process below: (your developed solutions **should not be limited** by these process):
 - **Ticket purchasing data** should be stored, including the transaction date, employee who are in charge, number of ticket sold for both adults and children should be known respectively
 - **Games and Attraction data**, able to indicate, when the game / attraction are built for the first time, age of the game / attraction since been build, player capacity, and also the duration of the game / attraction.
 - **Attendance data**, as it will be used to calculate the employee salary, the database should be able to store and indicate, what time the clock-in / clock-out time, are the employee late or coming on-time, and also indicate which employee are having overtime.
 - **Employee**, ensure the designed ERD can indicate the detail data of employee
 - **Member**, ensure the designed ERD can indicate the detail data of member

- c. Ensure the relationship between the entities in the diagram is connected properly.
- d. Ensure the entire component used to illustrate the diagram is using the appropriate component, referring to ERD that created using table notation.

2. Physical Model of the Database

- a. Ensure all the entity in ERD is translated into a database object
- b. Ensure all the relation between the translated database object is connected properly
- c. Ensure the database are having **minimum 3 constraint developed**, with following constraint are essentials in the system:

- Every games and attraction should be identified by ID, using the following format:
IND[999]-[G | A] where:
 - 1. **[999]**should be filled with the game / attraction sequence list.
 - 2. **[G | A]**indicates whether it is ID for Games or Attraction

Example: **IND001-G** indicates, Games number one in sequence list, **IND001-A** indicates, Attraction number one in the sequence list

- Number of ticket sold for each transaction, **cannot be lower than 0** for both adult and children tickets.
- Salary calculation **cannot be lower than 0**.
- d. Develop **minimum 5 data sample for every entity** designed to give a clear example of data stored in the database

Session 2: Create Master and Attendance Form

On the second phase of the information system development project, you have to make a good interface design. In this phase you are required to design the user interface of the information system to support Pemuda Theme Park business process above with the following criteria:

- a. Ensure the entire user interface developed, is concerning the 8 Golden Rules of User Interface Design, which is:
 - i. Strive for consistency
 - ii. Enable frequent users to use shortcuts, develop a minimum 5 shortcut to open designed form to help the frequent user
 - iii. Offer informative feedback, ensure there's a minimum 5 feedback to responds the user action
 - iv. Design dialog to yield closure
 - v. Offer simple error handling
 - vi. Permit easy reversal of actions
 - vii. Support internal locus of control
 - viii. Reduce short-term memory load, a simple breadcrumb navigation to help the user remember the form steps
- b. Clear and neat layout or component positioning is a key points to be concerned when developing the user interface

Ensure all the information fields on the user interface are using the proper component. E.g. using a radio button for gender fields.

There are 3 master forms that you need to build. These master forms will be divided into member, staff, and game. Each of these forms is able to view the list of the data also to insert new data, update data, and delete data. Below are the details criteria:

1. Member Form

- a. The form must show the list of the members.
- b. The list can be sorted by member name and registered date.
- c. The form support searching feature to find specific member by name.

- d. The specific member can be updated. When updating is on progress, the id and registered date will not be changed.
- e. The specific member can extend their membership with specific membership plan.
- f. The form also must be able to insert a new member with specific membership plan.
- g. Registered date on the detail list will be shown in format like "*Day, Date Month Year*". Example: Saturday, 12 September 2010.
- h. Validate member code must have format *[yyyy][xxxx]*, ex: *20150001*

2. Employee Form

- a. The form must show the list of the employees.
- b. The list can be sorted by employee name and shift.
- c. The form support searching feature to find specific employeeby name.
- d. The specific employee can be updated.
- e. Employee data can be inserted.

3. Games Form

- a. The form must show the list of the games.
- b. The list can be sorted by game name, duration, bought date, and status.
- c. This form support searching feature to find specific game.
- d. The specific game can be updated.
- e. Game data can be inserted.
- f. Bought date on the detail list will be shown in format like "*Day, Date Month Year*". Example: Saturday, 12 September 2010.

4. Attendance Form

- a. Attendance data of each employee either clock-in, clock-out and date can be inserted.Each attendance will be check whether it is late or on-time.(e.g. "Late", "On-time", or "Absent").
- b. Attendance data from previous day until today can be inserted. But, the system can't insert attendance data for the next day.
- c. Validate the admin cannot insert same employee in same date.
- d. The system can show the list of the attendance data on specific date which chosen by user.

Session 3: Create transaction form

In this session you must create 2 transaction form (selling ticket and view salary report). Selling ticket form is the form to help cashier input the ticket transaction to database. View salary report form is the form to help owner to see staff salary every month. You must follow this criteria:

1. Selling ticket form

- a. The form must show the clock in format “**hh:mm:ss dd/mm/yy**”.
- b. The cashier can select adult or child tickets for items.
- c. Validate the minimum quantity of ticket is 1.
- d. The transaction can choose member or nonmember.
- e. When the admin choose member, the program must follow this criteria:
 - The admin must input member code.
 - Validate the member code must valid and not expired.
 - Validate the member can get 1 free ticket (adult/child) in one day.
- f. When the admin choose nonmember, the program must follow this criteria:
 - The cashier can input the quantity of ticket.
 - The system will automatic using the price accordance with the day.
 - The transaction can use 3 types of payment (cash, debit, and credit card).
 - When payment using cash, then the cashier must input the nominal money. After that, the system will show the change.
 - When payment using debit and credit card, then the cashier must input code transaction and code of debit or credit card in format “**9999-9999-9999-9999**”.
- g. After the cashier input all data and click button submit, insert all data transaction to database.

2. Staff schedule form

- a. The form can view, add or edit the schedule for every staff.
- b. Each the schedule must have staff, games/attraction, shift and date.
- c. Validate the staff can't be allocated more than 1 games or attractions at the same shift and date.
- d. validate each game must have maximum 3 staffs
- e. When the staff want to leave, the admin can replace the staff with another staff that available in that date (substitute the staff).
- f. Only upcoming schedule that can be updated by admin.

- g. When the staff is replaced, the attendance of the staff become absent in that date

3. View salary report form

- a. The owner can select the month and year to view salary report of the chosen month. Validate the owner can select the month and year before now.
- b. When the user selected the month and year, the form will show all data staff (id, name, total attendance, total lateness, total overtime, and salary) and the summary of the selected month. The summary of the selected month must contains:
- Total salary all staff
 - Count lateness all staff
 - Count all staff
- c. When the user selected the staff, the form will show attendance report of the selected staff from the chosen month.

4. View selling ticket report form

- a. The owner can select the month or year to view selling ticket report from the chosen month. Validate the owner can select the month or year before now.
- b. When the user selected the month, the form will show all data transaction (date, quantity ticket adult, quantity ticket child, and quantity ticket member).